Inference II: the Bootstrap and Prediction

Applied Linear Models I • Lecture 5 of 16

Applied Linear Models I

Today

- The **bootstrap**: inference without a normality assumption.
- Two flavors: pairs (case) vs residual.
- Where it matters: **heteroskedasticity**.
- **Prediction** intervals vs confidence intervals – the irreducible "+1".

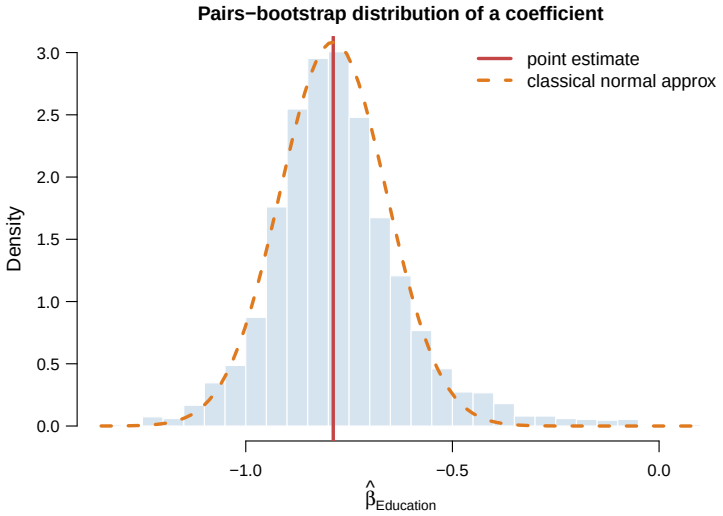
The bootstrap idea

Estimate the sampling distribution of $\hat{\beta}$ by **resampling the data**, rather than deriving it from assumptions.

- **Pairs bootstrap**: resample rows (x_i, y_i), refit. No assumption on the error variance – valid under heteroskedasticity.
- **Residual bootstrap**: fix X, resample residuals, $y_i^* = \hat{y}_i + e_i^*$, refit. Assumes *homoskedastic* errors.

Percentiles of the refit estimates $\Rightarrow$ confidence interval.

Bootstrap distribution (swiss)



Where the bootstrap earns its keep

Simulate **strong heteroskedasticity** (SD $\propto x^2$); measure *actual* coverage of nominal-95% intervals:

Interval	Empirical coverage
Classical	$\approx 88\%$ (under-covers)
Pairs bootstrap	$\approx 91\%$ (closer)

- Classical SEs assume constant variance – false here $\Rightarrow$ under-coverage.
- Pairs bootstrap never assumed it $\Rightarrow$ more honest (not perfect at small n).
- Complements HC3 robust SEs (Lecture 6).

Confidence vs prediction intervals

At a query point x_0:

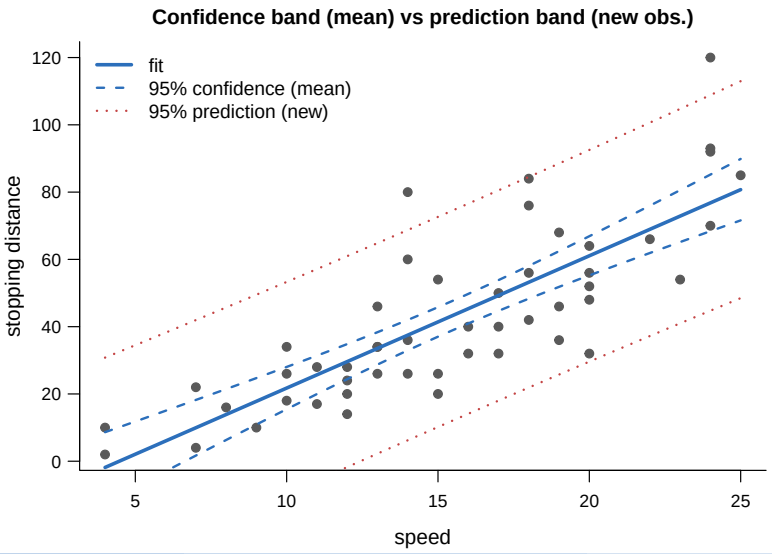
$$\text{mean: } \hat{\mu}_0 \pm t\,\hat{\sigma}\sqrt{x_0^\top (X^\top X)^{-1} x_0}$$

$$\text{new obs.: } \hat{\mu}_0 \pm t\,\hat{\sigma}\sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}$$

The "$+1$" is the irreducible variance of a single new outcome.

- Prediction interval is **always wider**.
- As $n \to \infty$: CI $\to$ a point; PI $\to \pm z\sigma$ (never vanishes).
- Monte Carlo: 95% PI covers a new outcome $\approx$ 95% of the time.

Confidence band vs prediction band



Summary

- Bootstrap estimates the sampling distribution by resampling; no normality needed.
- Pairs bootstrap is robust to heteroskedasticity; residual bootstrap is not.
- Prediction intervals add the new-obs. variance and never shrink to a point.
- Uncertainty explodes away from the data – beware extrapolation.

Next

Lecture 6: checking the assumptions – residual diagnostics and remedies.